

Question(s) for rec07 - Operator Overloading

7. Consider the following code. What would be the result when attempting to compile-and-run the program? Completely fill the circle next to your choice.

```
class Thing {
    friend ostream& operator<<(ostream& os, const Thing& rhs) {
        os << "Thing: " << rhs.val;
        return os;
    }
public:
    Thing(int val) : val(val) {}
    void setVal(int val) {
        this->val = val;                // Line A
    }
    Thing operator+=(const Thing& rhs) { // Line B
        val += rhs.val;
        return *this;
    }
private:
    int val;
};

int main() {
    Thing felix(17), heathcliffe(25);    // Line C
    (heathcliffe += felix).setVal(28);    // Line D

    cout << heathcliffe << endl;
}
```

- | | |
|---|---|
| ☐ Displays: Thing: 17 | ☐ Undefined |
| ☐ Displays: Thing: 25 | ☐ Compilation error at line A |
| ☐ Displays: Thing: 28 | ☐ Compilation error at line B |
| ☐ Displays: Thing: 42 | ☐ Compilation error at line C |
| ☐ Displays: Thing: | ☐ Compilation error at line D |
| ☐ Displays a blank line | ☐ None of the above |